

Ovarian Morphological and Functional Changes in Reproductively Senescent Hamsters^{1,2}

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ABSTRACT Ovaries from young and senescent hamsters were examined morphologically on days 8, 12 and 14 of pregnancy to determine if there were any differences between the two age groups. The senescent hamsters had fewer follicles present than the young on all days of gestation. The senescent females' corpora lutea experienced the greatest growth between days 8 to 12, whereas corpora lutea from young animals grew the most between days 12 and 14. Corpora lutea in senescent females did not grow at all between days 12 and 14. The lower number of follicles in senescent females was not due to the lower rate of implantation. Superovulation with PMS revealed that the senescent ovary was refractory.

It was concluded that there are quantitative not qualitative differences between ovaries from young and senescent hamsters during pregnancy. The refractory ovary of senescent females could be contributing to pregnancy wastage in this species by secreting less progesterone.

The litter size of female hamsters declines after 14 months (Soderwall, Kent, Turbyfill and Britenbaker, '60) due to reduced implantation and increased resorption (Thornycroft and Soderwall, '69). The cause(s) of the declining litter size is not at all clear. Ortiz ('55) reported that age changes in the uterus of the hamster could be reversed with PMS; she concluded that senescence of the reproductive system was due to pituitary hypofunction as the ovaries of senescent females appeared responsive to the PMS. Blaha ('64b) reported ova from senescent female hamsters were less viable than those from young females. He observed a decreased decidual cell reaction in pregnant senescent hamsters ('64a), and reported a deciduoma was much more difficult to induce in senescent females ('67); PMS, however, enhanced the inducibility of a deciduoma in aged hamsters (Blaha, '67). Blaha concluded that the aged hamster uterus was refractory to ovarian hormones, and reproductive failure with age in this species was a consequence of the uterine refractoriness and the less viable ova from senescent females.

Greenwald ('64) has shown that in young hamsters the ovary contains 13 non atretic antral follicles on day 8, and the number doubles to 26 by day 12. The doubling is preceded by a depletion of

pituitary FSH, which is required along with LTH to maintain pregnancy and presumably a functional corpus luteum in this species (Greenwald, Keever and Grady, '67 and Greenwald, '67). Hypofunction of either the pituitary or the ovary during pregnancy in senescent hamsters would probably result in the development of fewer follicles. As previous reports as to the causes of reproductive decline in the hamster conflict, ovaries from young and senescent pregnant hamsters were examined to determine if the aged group developed fewer follicles.

METHODS

1. Ovarian morphology

Young (3-6 months) and senescent (14-19 months) hamsters (*Mesocricetus auratus*) were bred with young males (3-6 months) and sacrificed at 2000 hours on days 8, 12 and 14 of pregnancy. The ovaries were removed, fixed for 24 hours in Bouin's solution, serially sectioned at 10 μ

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² Some preliminary results of this study have appeared in abstract form (Thornycroft, I. H., and A. L. Soderwall 1968 Fed. Proc., 27: 739).

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and stained with hematoxylin and eosin. The number and diameter of all antral follicles (atretic and non atretic) were determined. The follicular diameter measurement was made according to Greenwald ('64), and only atretic follicles larger than $235\ \mu$ were recorded. Smaller follicles are difficult to measure as the granulosa cells have almost disappeared, and these follicles probably do not represent very recent atresia (see below). The diameter of all corpora lutea in both age groups was also measured. The corpora lutea diameters were calculated according to the following formula:

$$D = \frac{1}{2} (D_1 + ((d_1 + d_2)/2)W)$$

where D is the final diameter of the gland; D_1 is its length in micra and is equal to the number of sections in which it is present, multiplied by $10\ \mu$ (thickness of each section); d_1 and d_2 are diameters, in ocular micrometer units, at the center of the corpus luteum and at right angles to each other; and W is the constant to convert ocular micrometer units taken at 50 power into micra.

2. Effect of reducing the number of implants on ovarian morphology

To ascertain if the reduced number of implants in senescent females could have an effect on ovarian morphology, young hamsters were unilaterally-salpingectomized under Nembutol anesthesia (10.0 mg per animal) on day 1 of pregnancy, and sacrificed at 2000 hours on days 8 and 14. Animals treated in this manner should possess one half the number of implants present in young intact females. At sacrifice the ovaries were removed and treated in the same manner as described above. This group will be referred to as the salpingectomized animals. In this paper, day 1 of pregnancy is the day following estrus.

3. Follicle turnover

To determine the approximate rate of follicular turnover, all atretic follicles larger than $300\ \mu$ were considered to have started atresia 48 hours or less, prior to sacrifice. The $300\ \mu$ figure was used as sections of an ovary taken from a cyclic animal 24 hours after ovulation revealed the largest atretic follicle to be $266\ \mu$.

These atretic follicles were probably those which began atresia sometime after 0900 hours on day 3 of the previous cycle (Greenwald, '61), thus they had been undergoing atresia for between 41 and 65 hours.

4. Superovulation

In another series of experiments, young and senescent cyclic females were superovulated. Each age group was split into two subgroups. One received a single subcutaneous injection of 30 IU of PMS (Pregnant Mare Serum, Equinex, donated by Ayerst Labs) at 1200 hours on day 1 of their estrous cycle (day of vaginal discharge) and the other received 60 IU at the same time. All animals were sacrificed at 0800 hours on day 2 of the next cycle. This procedure was found by Greenwald ('62) to achieve a maximum response with either dose. In addition 25 IU of HCG was given subcutaneously to all hamsters at 1900 hours on day 4 to insure complete ovulation. All animals were kept in a room which was illuminated for 14 hours, the lights being turned on at 0500 and off at 1900 hours. All animals were checked through at least three estrous cycles prior to use, by the lordosis response and confirmed by the post ovulatory vaginal discharge.

All tests of statistical significance in all experiments reported here were performed using the analysis of variance.

RESULTS

1. Follicular size and number

Senescent pregnant females had statistically fewer non atretic follicles than young animals on day 8 ($p < 0.01$), 12 ($p < 0.01$), and 14 ($p < 0.05$) of pregnancy (table 1). On days 8, 12 and 14 senescent ovaries contained 66.5, 61.3 and 66.7% respectively of the number of non atretic follicles contained in young ovaries. In both groups the number of non atretic follicles doubled between days 8 and 12 but remained constant between days 12 and 14. The number of atretic follicles present on day 8 in young and senescent animals did not differ statistically, but the senescent group had fewer present on day 12 ($p < 0.05$) and 14 ($p < 0.01$; table 1). The

TABLE 1
Number of antral follicles (Mean $\pm$ S.E.)

Day of pregnancy	Non atretic			Atretic		
	Young	Senescent	Salpingectomized	Young	Senescent	Salpingectomized
8	14.9 $\pm$ 1.1 (8) ¹	9.9 $\pm$ 0.9 (7)	12.9 $\pm$ 1.3 (8)	16.9 $\pm$ 1.8 (8)	13.0 $\pm$ 1.4 (7)	16.3 $\pm$ 1.6 (8)
12	27.1 $\pm$ 1.6 (8)	16.6 $\pm$ 2.7 (8)		26.4 $\pm$ 1.9 (8)	19.1 $\pm$ 2.4 (8)	
14	25.9 $\pm$ 2.9 (8)	17.3 $\pm$ 2.9 (8)	24.6 $\pm$ 1.7 (8)	40.0 $\pm$ 1.9 (8)	26.2 $\pm$ 1.6 (6)	38.6 $\pm$ 1.6 (8)

¹ Number of ovaries in ().

number of atretic follicles present on any day of pregnancy was greater than the preceding day studied in both age groups. The follicular sizes did not differ greatly between the two age groups (table 2).

In young animals on day 12 there were 23.4 atretic follicles larger than 300 μ , and in senescent 16.7. These follicles probably represent those follicles which became atretic between days 8 and 12. On day 14 there were 31.2 atretic follicles greater than 300 μ in young animals and 23.6 in the senescent, although the number of non atretic follicles remained constant between days 12 and 14 in both age groups.

2. Corpora lutea size and number

The corpora lutea in senescent females were slightly smaller than the corpora lutea in young animals on day 8 ($p < 0.01$). Between days 8 and 12 the corpora lutea in senescent animals grew more than those from young animals and were larger on day 12 ($p < 0.01$). The corpora lutea of young hamsters grew considerably between days 12 and 14, but those in senescent animals did not grow at all. On day 14 the young animals had larger corpora lutea than the senescent ($p < 0.01$).

The salpingectomized animals' corpora lutea were the same size as those taken from young animals on day 8, but were smaller on day 14 ($p < 0.01$); they had, however, somewhat larger corpora lutea than the senescent group on day 14 ($p < 0.01$; table 3).

3. Superovulation

There was no significant difference between the left and right ovary's response

to either of the doses of PMS, and there was no difference between the 30 and 60 IU dose responses within either age group. Accordingly, the results from the left and right ovary and from the 30 and 60 IU doses were combined within each age group. The young group, in response to PMS, ovulated 18.2 ± 1.3 ova per ovary and the senescent 13.5 ± 1.4 (Mean $\pm$ Standard Error). The difference between the two age groups was significant ($p < 0.05$). There were 18 ovaries in the senescent group and 20 in the young. Senescent animals ovulated 74.3 per cent as many ova as young animals in response to PMS.

DISCUSSION

On all days studied, ovaries from senescent hamsters contained fewer non atretic follicles than those from young animals. As FSH is part of the requirement for the maintenance of pregnancy in this species (Greenwald, '67), any reduction in its secretion rate or in the ovary's response would most probably result in fewer follicles maturing. The results presented here would, therefore, indicate changes in the pituitary-ovarian axis of the senescent hamsters. The number of implants is less in senescent than young hamsters (Thornycroft and Soderwall, '69) and any reduction could conceivably result in fewer follicles. In other words, the amount of FSH released, may be directly proportional to the number of implants. However, the salpingectomized group, which had 5.6 versus the 12.8 implants observed in young controls, did not differ from the young intact group, and it must be assumed that litter

TABLE 2
Follicle diameters (Mean $\pm$ S.E.)

Day of pregnancy	Non atretic (in micra)			Atretic (in micra)		
	Young	Senescent	Salpingectomized	Young	Senescent	Salpingectomized
8	391 $\pm$ 8 (8) ¹	393 $\pm$ 12 (7)	363 $\pm$ 9 (8)	337 $\pm$ 7 (8)	365 $\pm$ 8 (7)	349 $\pm$ 7 (8)
12	403 $\pm$ 7 (8)	402 $\pm$ 9 (8)		446 $\pm$ 7 (8)	467 $\pm$ 10 (8)	
14	388 $\pm$ 6 (8)	383 $\pm$ 7 (8)	390 $\pm$ 7 (8)	426 $\pm$ 5 (8)	435 $\pm$ 8 (8)	414 $\pm$ 6 (8)

¹ Number of ovaries in ().

TABLE 3
Corpora lutea diameter (in micra)

Days of pregnancy	Mean $\pm$ S.E.		
	Young	Senescent	Salpingectomized
8	865 $\pm$ 9 (8, 66) ¹	823 $\pm$ 10 (7, 44)	863 $\pm$ 8 (6, 43)
12	949 $\pm$ 16 (8, 65)	1005 $\pm$ 10 (8, 60)	
14	1152 $\pm$ 11 (9, 61)	1016 $\pm$ 13 (6, 42)	1083 $\pm$ 19 (8, 66)

¹ Numbers in parentheses are the number of ovaries and corpora lutea respectively.

size had no effect on the number of follicles maturing.

Both the young and senescent age groups approximately doubled their number of non atretic follicles between days 8 and 12, and between days 12 and 14 the number remained constant. The results in young animals agree well with those previously reported by Greenwald ('64). The increase between days 8 and 12 in both age groups was due to new follicular maturation, not reduced atresia, as the number of atretic follicles present on day 12 was greater than on day 8 in both age groups. In addition the number of follicles, on day 12, greater than 300 μ , and which were assumed in this study to be new atresias, was 23.4 in young animals and 16.7 in senescent, indicating all follicles in both age groups present on day 8 had undergone atresia by day 12. Greenwald ('64) had hypothesized a continual maturation and atresia of follicles in young animals from his observations of atretic follicles of all sizes and stages of atresia

on all days of pregnancy. His conclusions are quantitatively supported here. Between days 12 and 14 where the number of non atretic follicles remained constant in both age groups, the number of atretic follicles greatly increased, again indicating a substantial follicular turnover in young and senescent animals. There were 31.2 atretic follicles greater than 300 μ in the young animals on day 14 and 23.6 in the senescent. These follicular results do not indicate any major ovarian dysfunction in senescent females, but rather a quantitative change. The ovaries from senescent females underwent the same qualitative changes as those from young females, but at a lower rate.

A lower plasma concentration of, or ovarian refractoriness to FSH could account for the results. The investigate the latter possibility, ovarian responsiveness to PMS was measured in both age groups; and the senescent ovulated 74.3% as many ova as young animals. The aged ovary was, therefore, refractory to PMS and presumably FSH activity.

Ovarian refractoriness could account entirely for the fewer follicles present in aged ovaries. The pituitary and plasma would, of course, have to be assayed for FSH activity on various days of pregnancy to determine if any pituitary hypofunction is also present. It is interesting to note, however, that senescent ovaries from pregnant animals contained approximately 65% as many ova as those from young females, whereas senescent females ovulated 73% as many ova in response to PMS as young animals. The difference between the two experiments could be explained by less

pituitary FSH synthesis or release. If there is less pituitary secretion it would have to be of secondary importance to the ovarian refractoriness, as the above discrepancy is small.

The smaller corpora lutea on day 14 and the absence of growth between days 12 and 14 in senescent animals is probably associated with this group's lower litter size. The reason being, the salpingectomized group had smaller corpora lutea than the young intact on day 14 but not on day 8, indicating that number of implants is related to corpora lutea size on day 14, and in addition the hamster placenta may be synthesizing a LTH like hormone. Greenwald ('67) concluded that the placenta may be synthesizing LTH as pituitary LTH is not required for the maintenance of pregnancy after day 12. Klein ('38) reported fetectomized hamster's corpora lutea did not regress, again indicating a LTH like compound may be synthesized by the hamster placenta. The corpora lutea results in young animals agree well with those previously reported (Greenwald, Keever and Grady, '67).

The much greater growth of the hamster corpus luteum between days 8 and 12 is difficult to explain as smaller corpora lutea would be anticipated from the follicular results. Greenwald ('67) has demonstrated LTH to structurally maintain the corpus luteum of pregnancy and the results from the senescent animals on day 12 may, therefore, be due to larger than normal levels of LTH. Excessively high titers of LTH are presumed to exist in senescent rats (Aschheim, '66). Peripheral levels of FSH and LTH will have to be assayed to explain the corpora lutea results in senescent hamsters.

From the results presented here, the following hypothesis of ovarian function in senescent hamsters is proposed. The ovary is refractory to FSH, explaining the reduced number of follicles present in ovaries from pregnant animals. Less FSH may also be synthesized or released by the pituitary, as there is a discrepancy between the superovulation results and the number of follicles present during gestation. The differences demonstrated between the ovaries from young and senescent hamsters are

quantitative not qualitative. Both groups experienced a continual maturation and atresia of follicles throughout pregnancy, however, the rate of follicular turnover was lower in senescent females.

Orsini and Meyer ('62) have reported low progesterone levels in young pregnant hamsters result in both resorption and fewer implants. It is, therefore, possible that the refractory ovary of senescent females contributes to the increased resorption and decreased implantation rates of senescent females (Thornycroft and Soderwall, '69) if ovaries from senescent females secrete less progesterone. Blaha ('68) has demonstrated that senescent females with ovaries transplanted from young females were more able to implant, had fewer resorptions and were more able to carry to term ova from young females than intact senescent females which received only ova from young animals. Blaha's results ('68) would, therefore, also indicate the ovary of senescent females to be refractory and contributing to pregnancy wastage.

Other factors are also involved in the decline of the reproductive performance of hamsters. Ortiz ('55) injected PMS into senescent hamsters and concluded that there was a pituitary hypofunction, although, her results could have been partially due to a refractory ovary. Blaha ('64a and b, and '67) has reported ova from senescent females are less viable than those from young animals, and the decidual cell reaction more difficult to induce and less extensive in the aged hamsters. He concluded that the decline in litter size with age was due to the less viable ovum and uterine refractoriness. His results could also have been partially due to a refractory ovary as large amounts of PMS increased the inducibility of a deciduoma in senescent hamsters.

In summary, the ovary of senescent female hamsters has been demonstrated to be refractory, and may consequently contribute to the failure of pregnancy in this species. However, a combination of factors is in all probability responsible for the decline in litter size with age. The ovum, pituitary, hypothalamus, ovary and uterus

all contributing. Which is the most critical, if any, awaits further elucidation.

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